Plumb — Developer Flow Diagrams

> **For:** Software developers implementing Plumb · **Version:** 1.0

>

> Ten flow diagrams showing the request flows, state machines, and data pipelines you will implement. Each diagram is paired with the architectural invariants that must hold in code. Colors follow `@plumb/tokens`: teal = control flow, amber = decisions/warnings/storage, dark = context.

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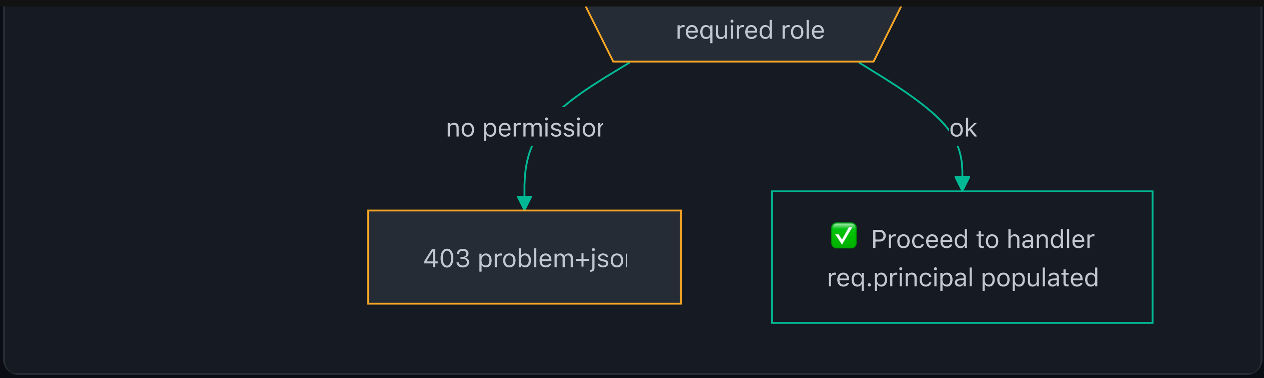
1. Authentication Flow (OIDC → JWT → RBAC)

What happens on every login: federated IdP → Cognito token → API Gateway validation → Redis-cached principal → per-route RBAC.

Developer note:
· @plumb/auth plugin does steps 6–9
· Idempotency-Key honored on mutating routes
· x-request-id propagated end-to-end
👤 User
(mobile or web)
Login attempt
IdP: Okta / Entra / Auth0
redirect w/ code
OIDC authorization cod
Exchange code → ID + access token
(Cognito user pool)
JWT: sub, email, org_id, role, tenant_id, exp
API Gateway validates:
· signature (JWKS)
· expiry
· audience
invalid
valid
401 problem+jsor
with trace_id
Principal cached in Redi
TTL 5 min
Service RBAC check
req.principal.role vs



Developer rules:

- Use the `@plumb/auth` plugin — it does steps 6–9 (JWT validate, principal attach, 401/403 problem+json). Don't reimplement.
- Every mutating route honors `Idempotency-Key`; every error carries `trace_id` (RFC 7807).
- Phase 2: service-to-service calls switch from bearer JWT to SPIFFE SVIDs (TTL ≤ 1 h) — the plugin interface stays the same.

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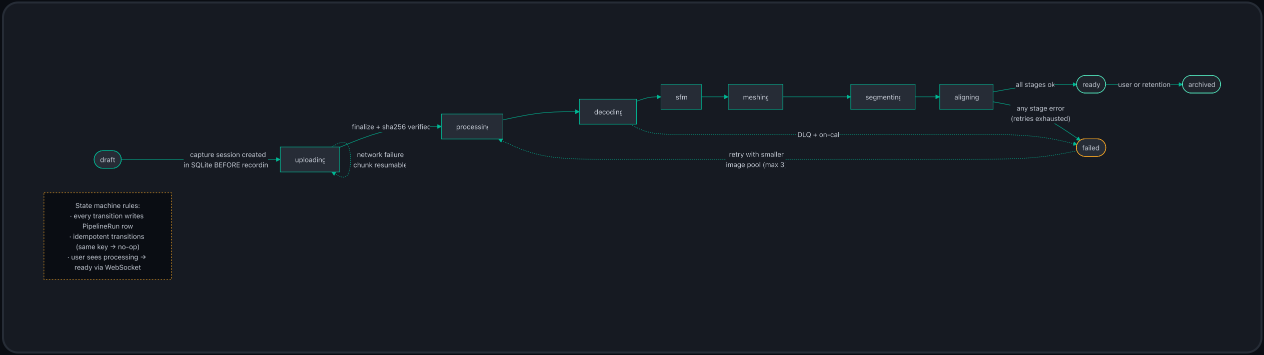
2. Capture Lifecycle State Machine

The capture goes draft → uploading → processing (5 sub-stages) → ready | failed | archived. This is the heart of the ingestion system.



Developer rules:

- `CaptureSession` row is inserted in SQLite **before** recording starts — crashes are recoverable.
- Every transition writes a `PipelineRun` row; transitions are idempotent (same key → no-op).
- Retry policy: SfM/MVS retry with a smaller image pool, max 3 attempts, then DLQ → on-call.
- User-visible state flips `processing → ready` via WebSocket push from realtime-gateway.

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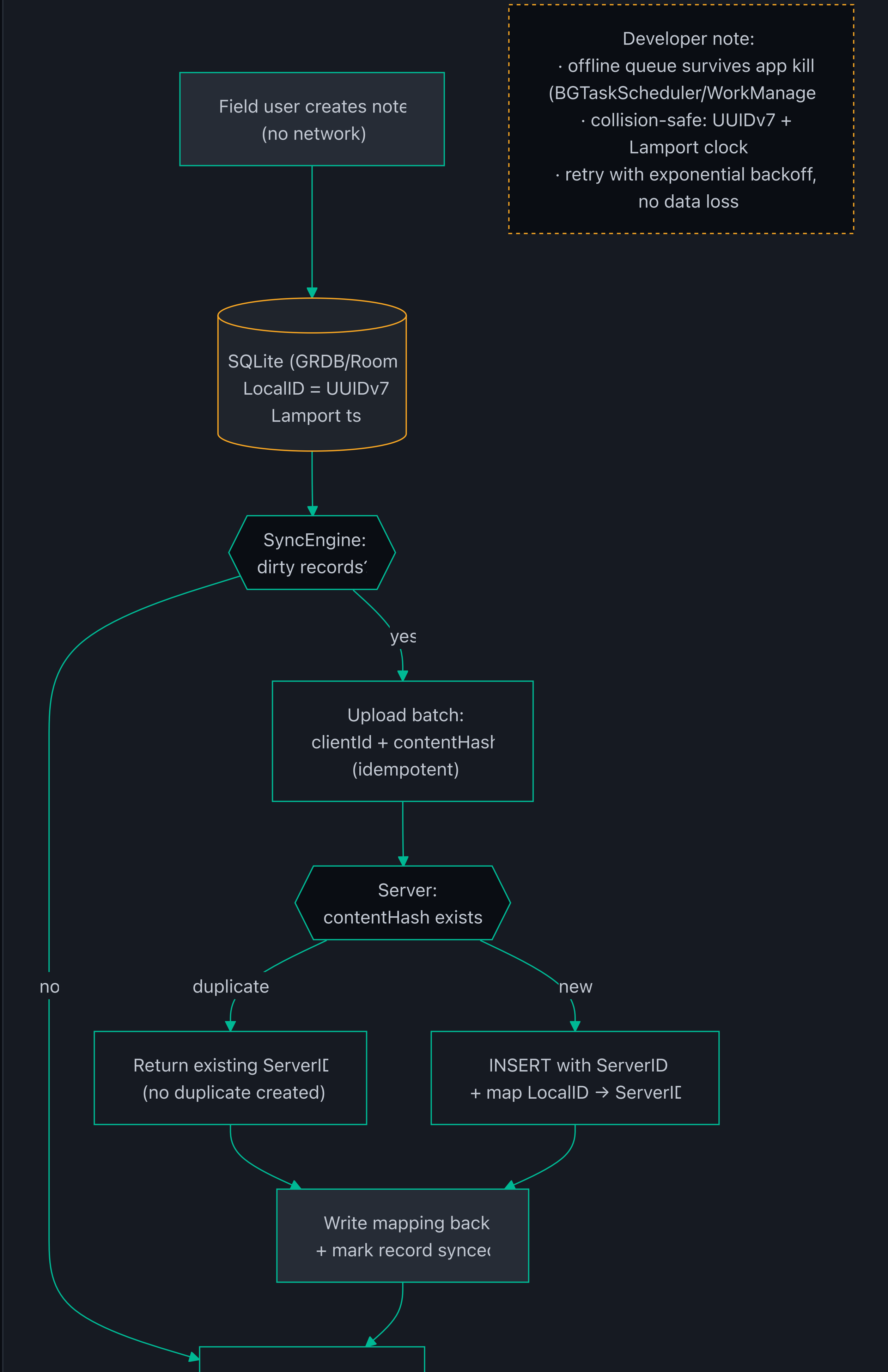
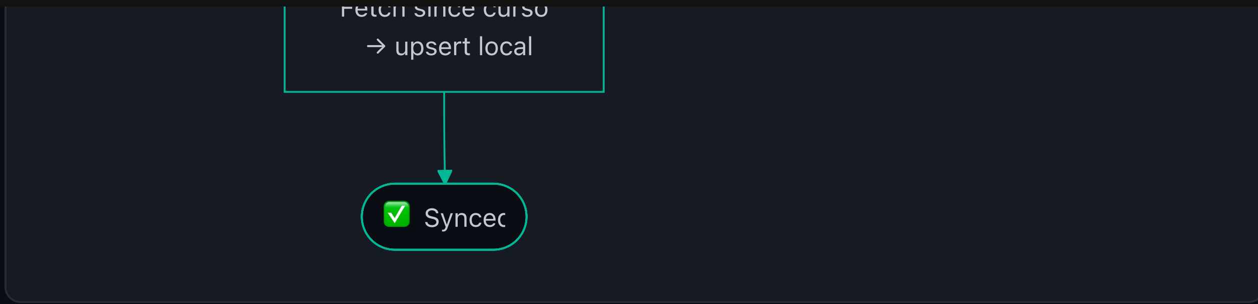
3. Offline-First Sync (Field Notes)

The field is offline ~40% of the day. The sync engine must never lose data and never duplicate.


Developer note:
· offline queue survives app kill
(BGTaskScheduler/WorkManage
· collision-safe: UUIDv7 +
Lamport clock
· retry with exponential backoff,
no data loss
Field user creates note
(no network)
SQLite (GRDB/Room
LocalID = UUIDv7
Lamport ts
SyncEngine:
dirty records'
yes
no
Upload batch:
clientId + contentHash
(idempotent)
Server:
contentHash exists
duplicate
new
Return existing ServerID
(no duplicate created)
INSERT with ServerID
+ map LocalID → ServerID
Write mapping back
+ mark record synced



Developer rules:

- LocalID = UUIDv7 (time-ordered, collision-safe); Lamport timestamps for ordering.
- Idempotency via `clientId + contentHash` — server returns existing ServerID on duplicate.
- Background upload via BGTaskScheduler (iOS) / WorkManager (Android); survives app kill and airplane mode.
- On reconnect: pull-by-cursor (since `lastSyncCursor`) → upsert locally.

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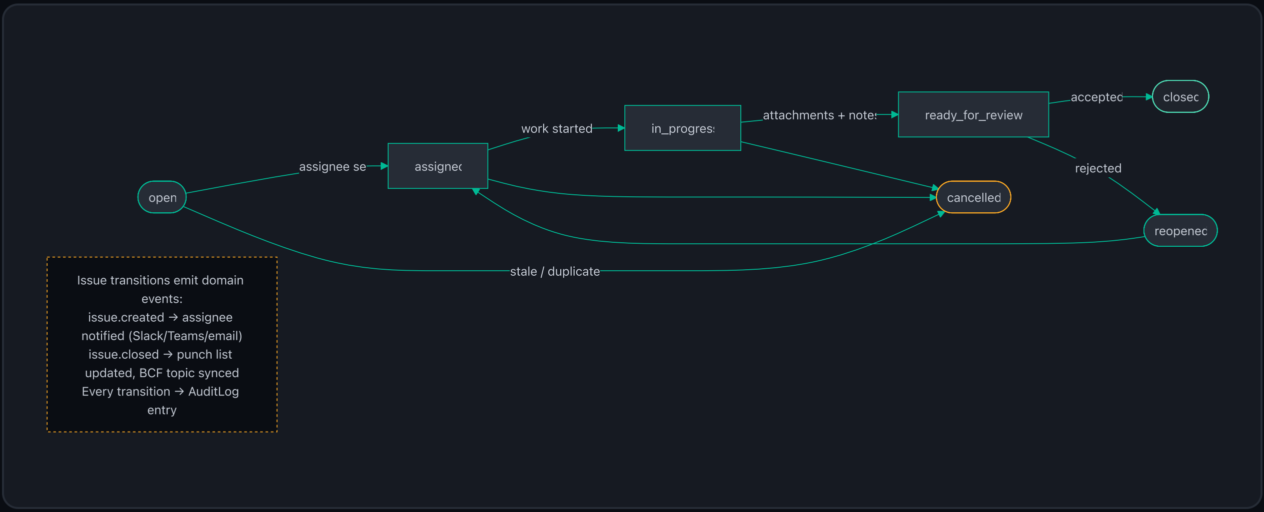
4. Issue Workflow State Machine

Issues flow open → assigned → in_progress → ready_for_review → closed, with reopen and cancel paths.



Developer rules:

- Every transition emits a domain event → notify (Slack/Teams/email) + AuditLog entry.
- BCF 3.0 topic sync on close (Plumb Model ↔ external BIM tools).
- Punch-list and progress rollups consume `issue.closed` events — never poll.

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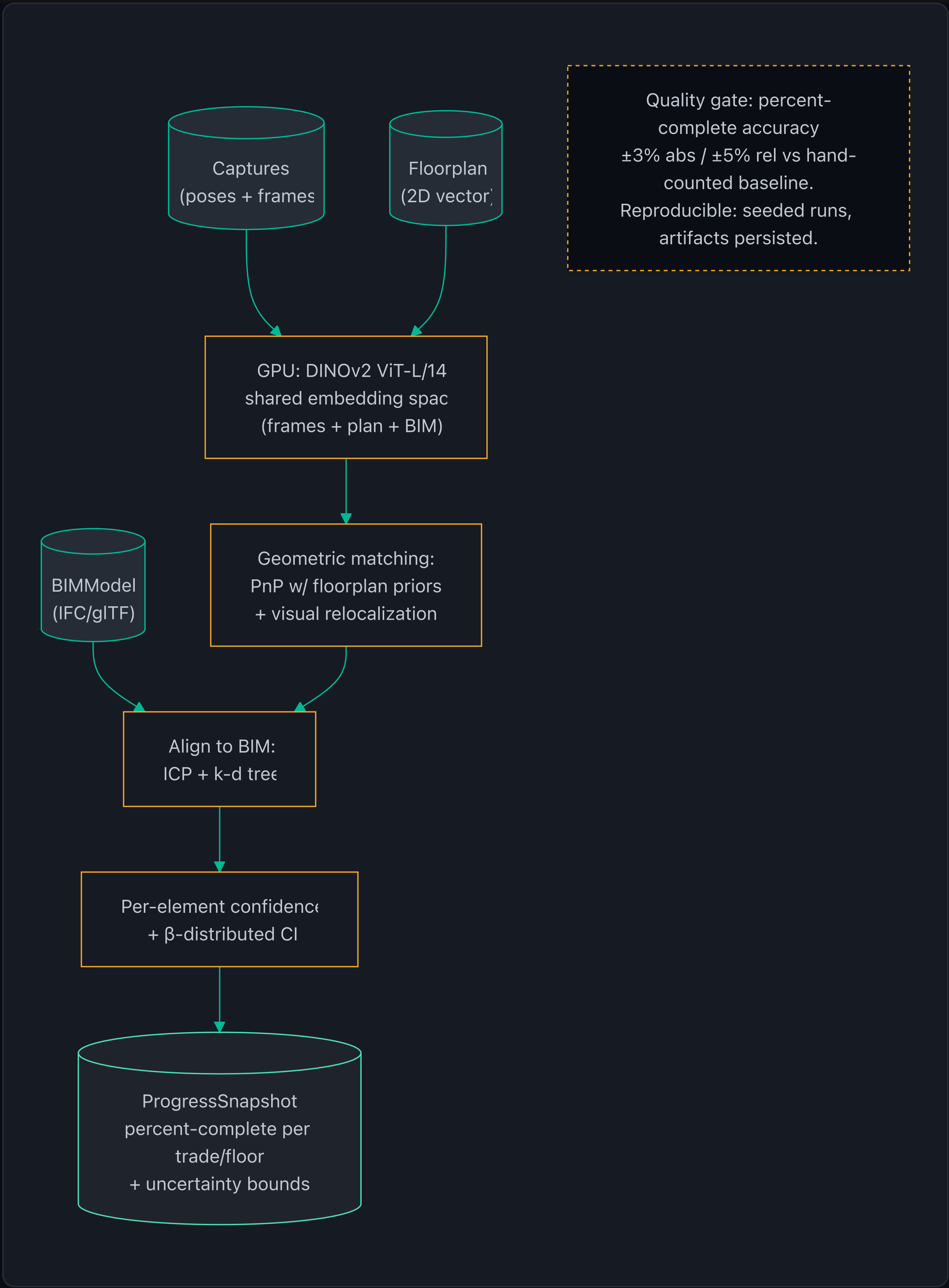
5. BIM-vs-Reality Alignment Pipeline

The math behind Plumb Track: how captures, floorplans, and BIM models fuse into per-trade percent-complete.



Developer rules:

- DINOv2 ViT-L/14 embeds frames, floorplans, and BIM in one space; PnP with floorplan priors recovers pose; ICP aligns to BIM mesh.

- Output carries β-distributed confidence intervals — never a bare number.
- Quality gate: ±3% abs / ±5% rel vs hand-counted baseline. Runs in CI on synthetic fixtures.
- Reproducible: seeded runs; every artifact persisted.

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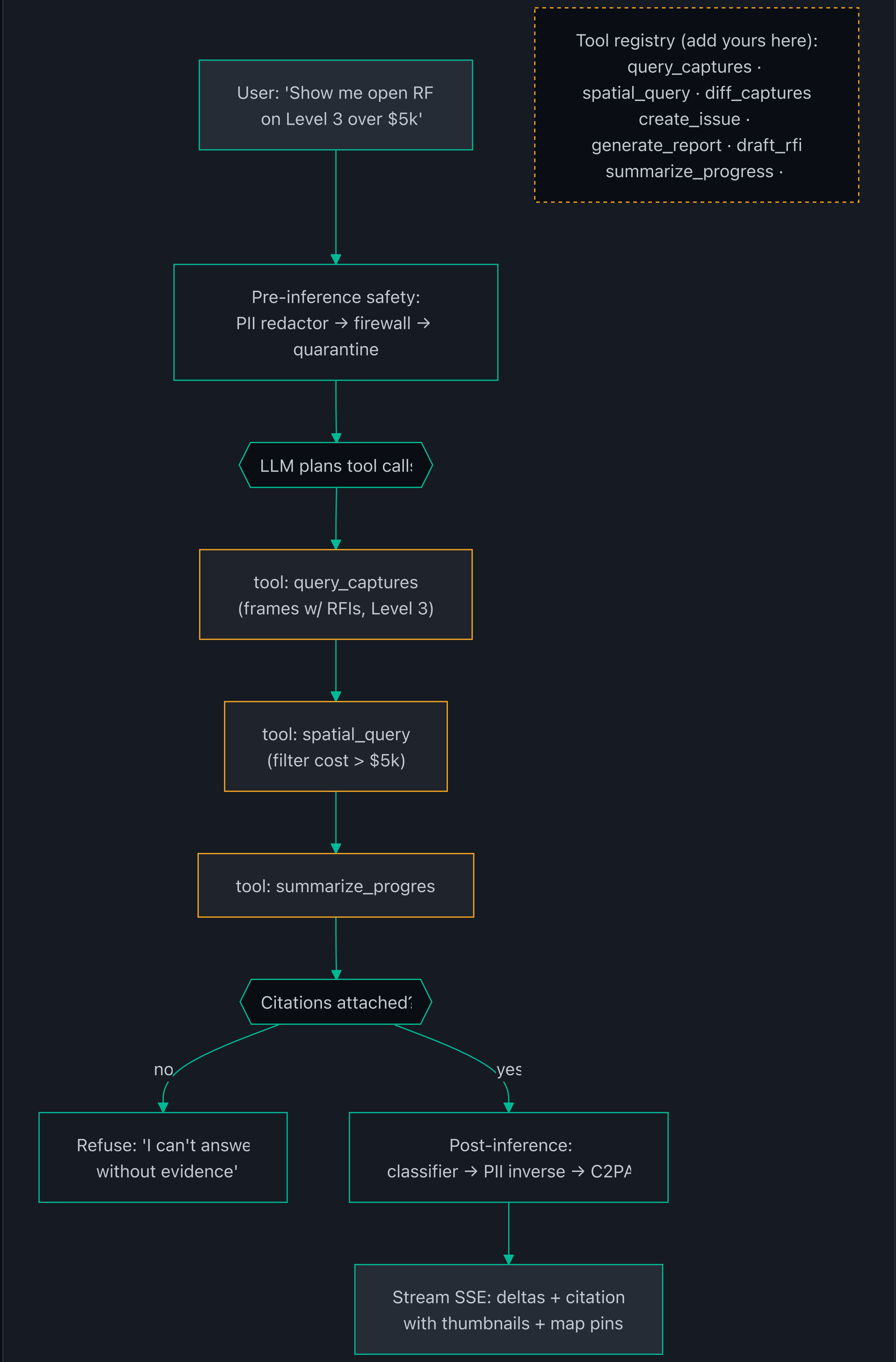
6. AI Copilot Function-Calling Loop

How Copilot answers "Show me open RFIs on Level 3 over $5k" — with citations and guardrails.


Tool registry (add yours here):
query_captures ·
spatial_query · diff_captures
create_issue ·
generate_report · draft_rfi
summarize_progress ·
User: 'Show me open RF
on Level 3 over $5k'
Pre-inference safety:
PII redactor → firewall →
quarantine
LLM plans tool call
tool: query_captures
(frames w/ RFIs, Level 3)
tool: spatial_query
(filter cost > $5k)
tool: summarize_progres
Citations attached
no
yes
Refuse: 'I can't answe
without evidence'
Post-inference:
classifier → PII inverse → C2P
Stream SSE: deltas + citation
with thumbnails + map pins

Developer rules:

- **Refusal-on-no-citation**: if the LLM's answer can't cite a project artifact, refuse. No exceptions.
- Tool registry is open — add tools in `services/copilot-service/src/tools/` (query_captures, spatial_query, diff_captures, create_issue, generate_report, draft_rfi, summarize_progress, measure).
- Pre-inference: PII redactor → prompt firewall → context quarantine. Post-inference: content classifier → PII inverse → C2PA provenance stamp.
- All inference is tenant-pinned (no cross-tenant context ever).

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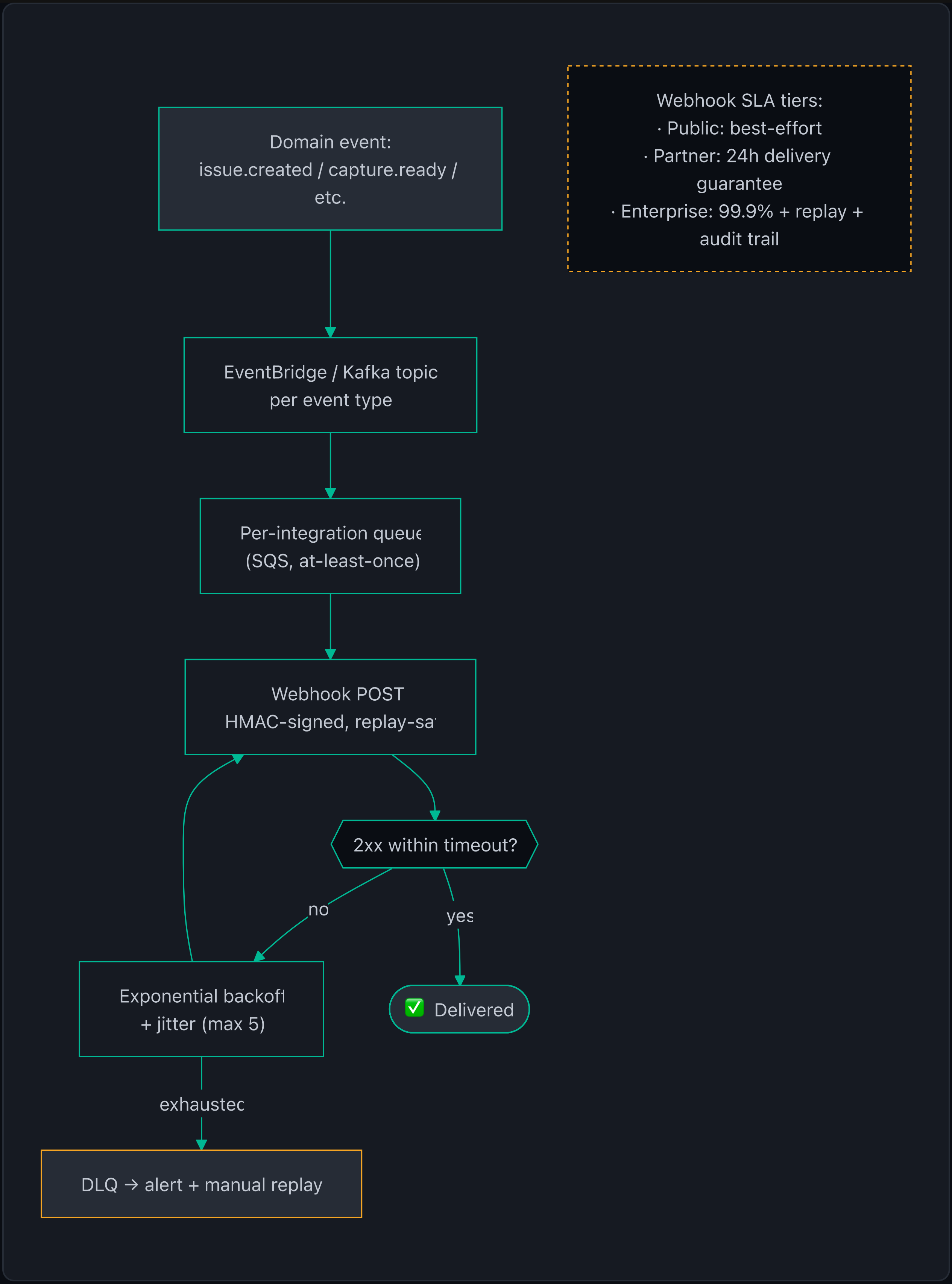
7. Webhook Delivery (Reliable, Replayable)

Every domain event can fan out to partner integrations. Delivery must be at-least-once with audit.



Developer rules:

- HMAC-signed payloads; signature header `X-Plumb-Signature`; replay via `X-Plumb-Event-Id` dedup.
- Exponential backoff + jitter, max 5 attempts, then DLQ + alert.

- SLA tiers: Public (best-effort) / Partner (24 h delivery) / Enterprise (99.9% + replay + audit trail).

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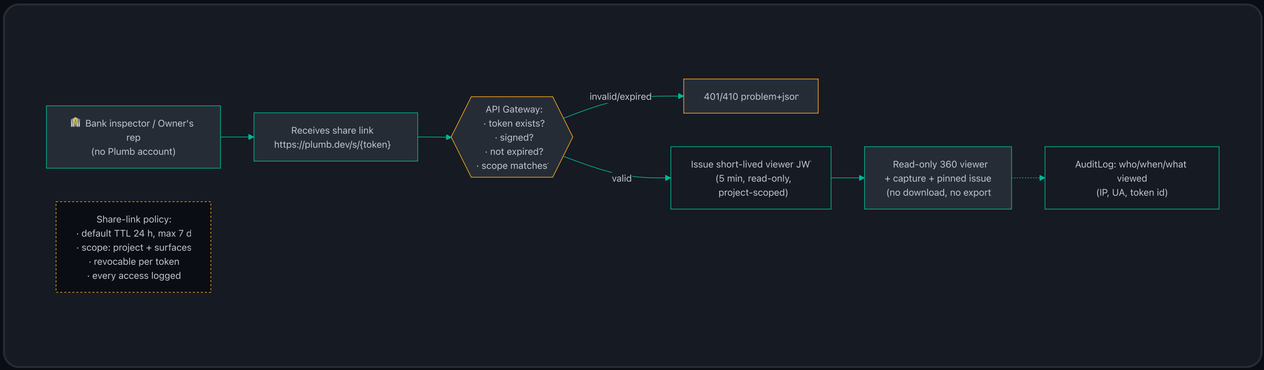
8. Share-Link Access (External Read-Only)

Owner's reps and bank inspectors have no account. Share links give scoped, expiring, revocable access.



Developer rules:

- Token = signed, scoped (project + surfaces), default TTL 24 h / max 7 d, revocable per-token.
- Exchange token → short-lived (5 min) read-only viewer JWT. No download, no export.
- Every access logged: token id, IP, UA, timestamp → AuditLog.

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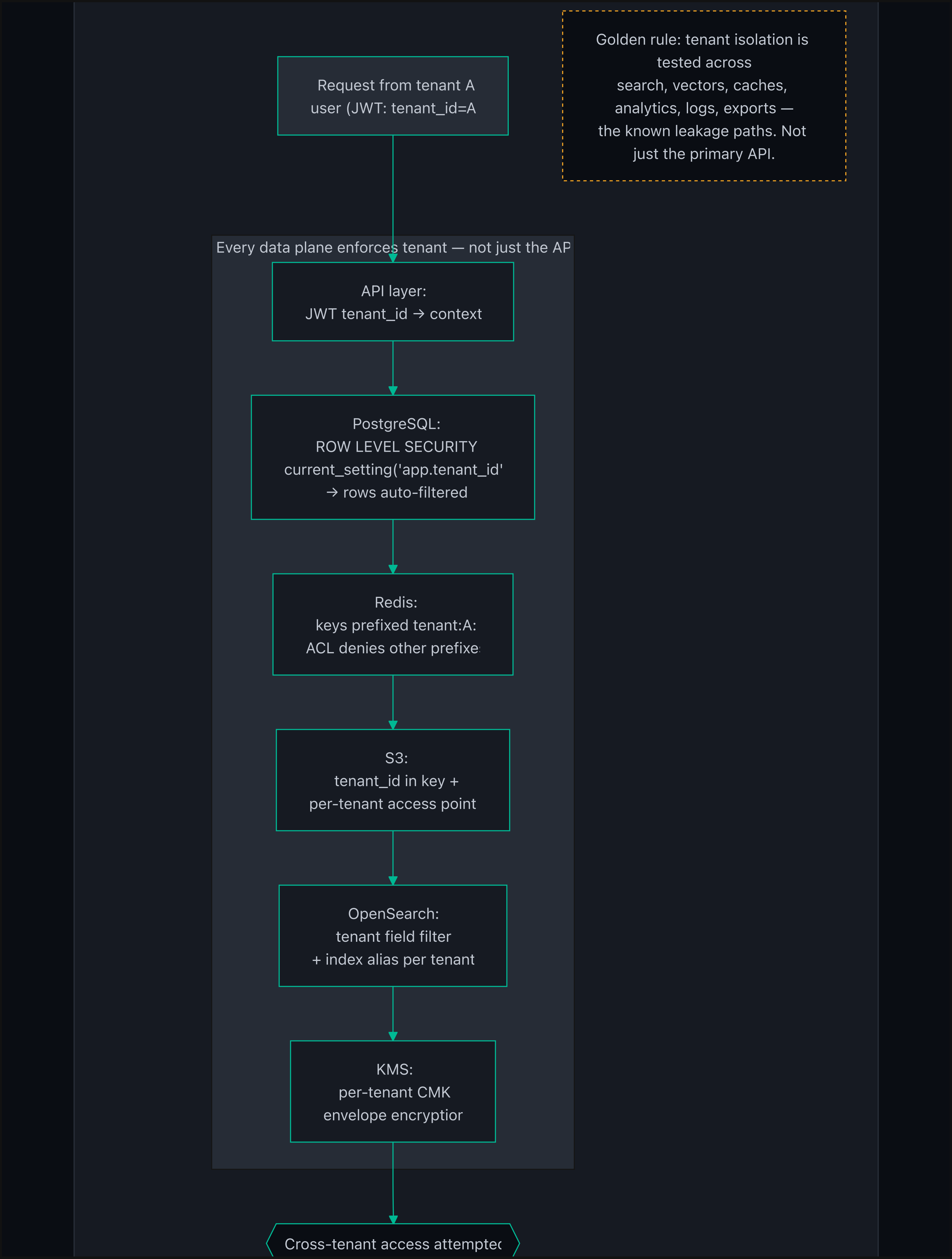
9. Tenant Isolation — Every Data Plane

Multi-tenancy is enforced at six layers, not just the API. This is the architecture's most important security invariant.



Golden rule: tenant isolation is tested across search, vectors, caches, analytics, logs, exports — the known leakage paths. Not just the primary API.
Request from tenant A user (JWT: tenant_id=A
Every data plane enforces tenant — not just the AP
API layer: JWT tenant_id → context
PostgreSQL: ROW LEVEL SECURITY current_setting('app.tenant_id' → rows auto-filtered
Redis: keys prefixed tenant:A: ACL denies other prefixe
S3: tenant_id in key + per-tenant access point
OpenSearch: tenant field filter + index alias per tenant
KMS: per-tenant CMK envelope encryptior
Cross-tenant access attemptec

Developer rules:

- Postgres: RLS on every table using `current_setting('app.tenant_id')` — set at connection pool checkout.
- Redis: keys prefixed `tenant::` + ACLs; S3: tenant in key + per-tenant access points; OpenSearch: tenant filter + alias; KMS: per-tenant CMK.
- **Test tenant isolation across search, vectors, caches, analytics, logs, and exports** — the known leakage paths. Not just the primary API.

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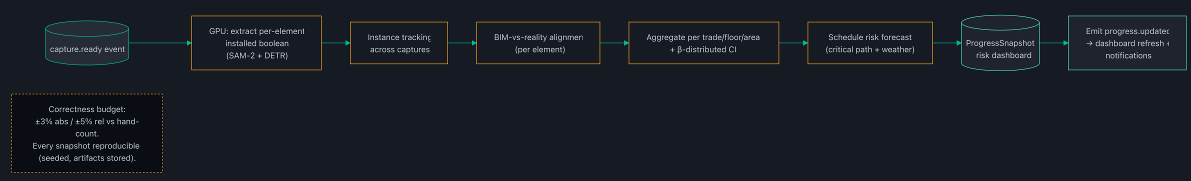
10. Progress Pipeline (Capture → Dashboard)

From `capture.ready` to the EV S-curve on the dashboard, in one event-driven flow.



Developer rules:

- Consume `capture.ready` events; never trigger on a schedule.
- Per-element installed booleans → trade/floor aggregation with CI → schedule risk (critical path + weather).
- Emit `progress.updated` → dashboard refresh + notifications. Snapshots are immutable + reproducible.

Implementation checklist for each flow

Flow	Package/service	Test seam	Verify with
1 Auth	`packages/auth`, `services/user-service`	`verifyToken` option	`pnpm --filter=@plumb/auth test`
2 Capture	`services/capture-service` (P1)	FakePgClient + Step Functions mock	`pnpm --filter=@plumb/capture-service test`
3 Sync	`apps/mobile-kmm` (P1)	JVM unit tests + mocked remote	`gradle :mobile-kmm:jvmTest`
4 Issues	`services/field-service` (P1)	Event capture on service	Event emitted assertions
5 Alignment	`services/imgproc-service` (P1)	Synthetic Blender fixtures	CI numerical gate ±3/±5%

| 6 Copilot | `services/copilot-service` (P2) | Red-team + hallucination eval | Citation-faithfulness suite |

| 7 Webhooks | `services/integration-hub` (P2) | Mock webhook endpoint | Retry/DLQ tests |

| 8 Share links | `services/user-service` | Token mint/verify unit tests | 401/410 on expired |

| 9 Tenancy | All services | Cross-tenant probe tests | Isolation suite in CI |

| 10 Progress | `services/track-service` (P2) | Synthetic captures → snapshots | Tolerance band checks |

End of developer flow document.